

WITTENBERG, Germany

WMO: 104740

Lat: 51.8892N

Lon: 12.6444E

Elev: 106

StdP: 100.06

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.0	-10.2	-16.7	0.9	-11.7	-13.6	1.2	-9.6	11.6	6.6	10.4	5.6	2.3	50	0.544

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	30.0	18.9	28.0	18.3	26.3	17.4	20.1	27.6	19.0	25.6	18.2	24.3	3.2	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	12.8	21.6	16.7	12.0	21.2	15.8	11.4	21.0	57.8	27.5	54.4	25.5	51.8	24.6	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 9.5	8.3	7.2	DB	-14.1	34.2	4.9	1.9	-17.6	35.6	-20.5	36.7	-23.2	37.8	-26.8	39.2	(4)
(5)			WB	-14.6	22.7	4.8	1.5	-18.0	23.7	-20.9	24.6	-23.6	25.4	-27.1	26.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.2	0.4	0.0	4.5	8.1	13.4	16.2	18.7	18.3	14.3	9.8	4.4	1.7
	DBStd	7.76	5.63	5.19	3.94	3.69	3.86	3.64	3.44	3.36	2.98	3.82	3.69	4.05
	HDD10.0	1324	299	281	173	81	13	0	0	0	2	51	169	256
	HDD18.3	3454	557	513	429	308	157	85	39	43	125	266	417	514
	CDD10.0	1037	1	1	3	22	119	187	269	258	132	44	2	1
	CDD18.3	126	0	0	0	0	5	22	49	43	6	1	0	0
	CDH23.3	1213	0	0	0	1	61	221	496	380	52	3	0	0
	CDH26.7	327	0	0	0	0	2	57	152	111	6	0	0	0
Wind	WSAvg	3.1	3.8	3.5	3.5	3.2	2.9	2.7	2.7	2.4	2.8	3.1	3.4	3.5
Precipitation	PrecAvg	614	51	38	44	32	57	59	74	61	52	40	52	52
	PrecMax	831	94	78	113	92	124	137	182	140	146	90	131	101
	PrecMin	427	4	9	8	1	10	17	11	23	11	8	1	18
	PrecStd	110	23	19	27	20	32	28	38	33	33	22	28	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.6	14.5	17.5	22.5	26.8	31.4	32.1	33.1	27.6	22.7	14.4	12.2
		MCWB	9.2	10.5	11.4	13.4	16.4	19.3	19.9	20.6	18.0	16.3	11.2	9.6
	2%	DB	9.3	11.1	14.2	19.2	25.1	28.1	30.2	29.4	24.7	19.5	12.3	10.6
		MCWB	7.5	8.4	9.9	12.0	15.6	18.2	18.9	19.0	16.8	14.3	10.3	8.5
	5%	DB	8.1	8.2	11.8	17.0	23.3	25.8	28.3	27.3	22.3	17.2	10.8	9.2
		MCWB	6.4	6.2	8.4	11.0	14.7	16.7	18.5	18.1	16.0	13.3	8.9	7.7
	10%	DB	6.8	6.3	10.1	14.8	21.0	23.7	26.2	25.3	19.9	15.4	9.5	7.5
		MCWB	5.2	4.5	7.3	9.7	13.8	16.2	17.6	17.4	14.8	12.0	7.8	6.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	10.8	12.1	14.0	18.2	20.2	22.2	21.7	18.8	16.8	11.9	9.7
		MCDB	11.2	14.0	15.9	20.5	24.6	28.6	30.2	31.1	25.2	21.3	13.5	11.5
	2%	WB	7.9	8.4	10.3	12.6	16.8	18.8	20.3	19.9	17.7	15.0	10.5	8.6
		MCDB	9.1	10.6	13.7	18.5	22.7	25.5	27.7	27.1	22.8	18.6	12.2	10.2
	5%	WB	6.6	6.5	8.8	11.3	15.5	17.9	19.1	19.0	16.7	13.6	9.2	7.7
		MCDB	7.9	8.2	11.1	16.0	21.2	24.0	25.7	25.3	20.7	16.8	10.7	9.2
	10%	WB	5.3	4.6	7.5	10.1	14.3	16.6	18.2	18.1	15.6	12.3	7.9	6.2
		MCDB	6.8	6.2	9.8	14.3	20.0	22.5	24.6	23.6	19.5	14.8	9.3	7.4
Mean Daily Temperature Range	5% DB	MDBR	4.8	6.1	7.3	9.1	10.2	9.6	10.4	10.4	8.9	7.8	5.4	4.4
		MCDBR	5.1	7.9	10.5	12.6	13.6	13.6	14.2	13.9	12.5	10.6	6.9	5.6
		MCWBR	4.5	5.9	7.0	6.9	6.5	6.1	6.1	6.1	6.2	6.4	5.5	4.6
	5% WB	MCDBR	5.3	7.8	9.6	11.6	12.3	12.1	12.6	12.1	11.2	9.9	6.8	5.2
		MCWBR	5.0	6.0	6.8	6.6	6.5	5.8	6.3	5.7	6.2	6.4	5.6	4.5
Clear-Sky Solar Irradiance	taub	0.318	0.341	0.382	0.415	0.416	0.412	0.432	0.425	0.387	0.365	0.341	0.312	
	taud	2.354	2.333	2.235	2.200	2.240	2.277	2.231	2.266	2.350	2.385	2.390	2.362	
	Ebn at Noon	681	769	800	818	837	843	817	801	791	731	646	621	
	Edh at Noon	62	83	111	129	130	127	131	120	99	79	61	53	
All-Sky Solar Radiation	RadAvg	0.69	1.38	2.39	3.96	4.87	5.19	4.89	4.22	3.06	1.74	0.82	0.52	
	RadStd	0.08	0.24	0.29	0.51	0.58	0.42	0.63	0.36	0.37	0.20	0.11	0.06	

Historical Trends

		Heating		Cooling		Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page